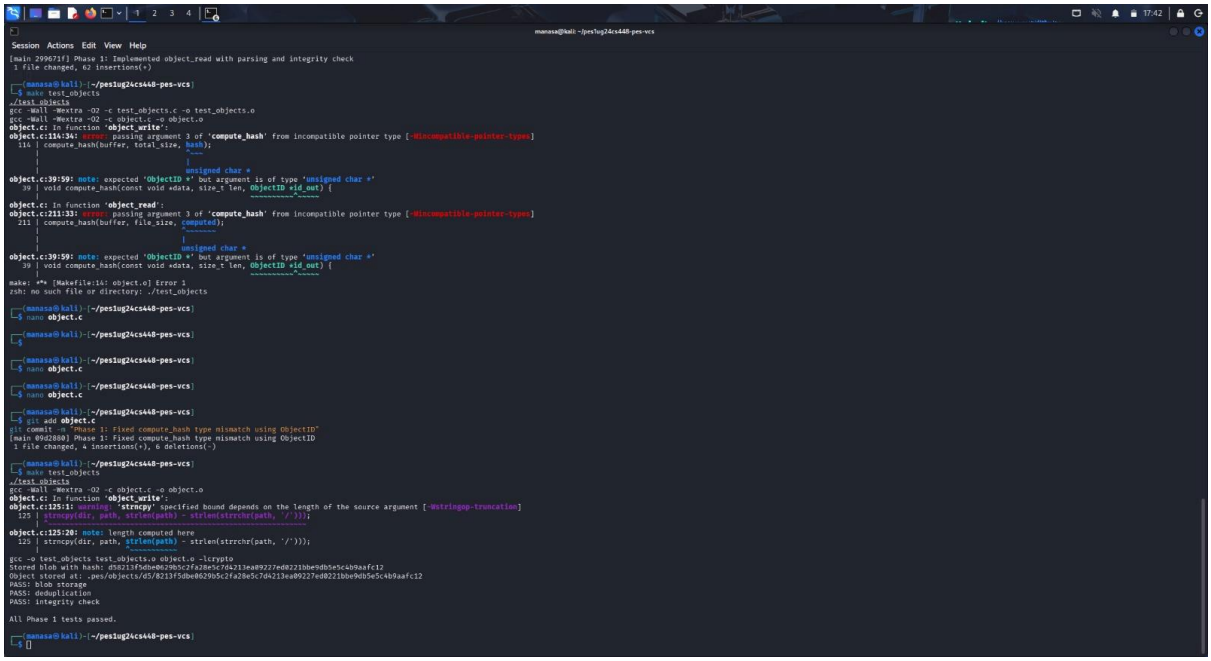
PES1UG24CS448

SHRIYA K

4H

SCREENSHOTS

1.a

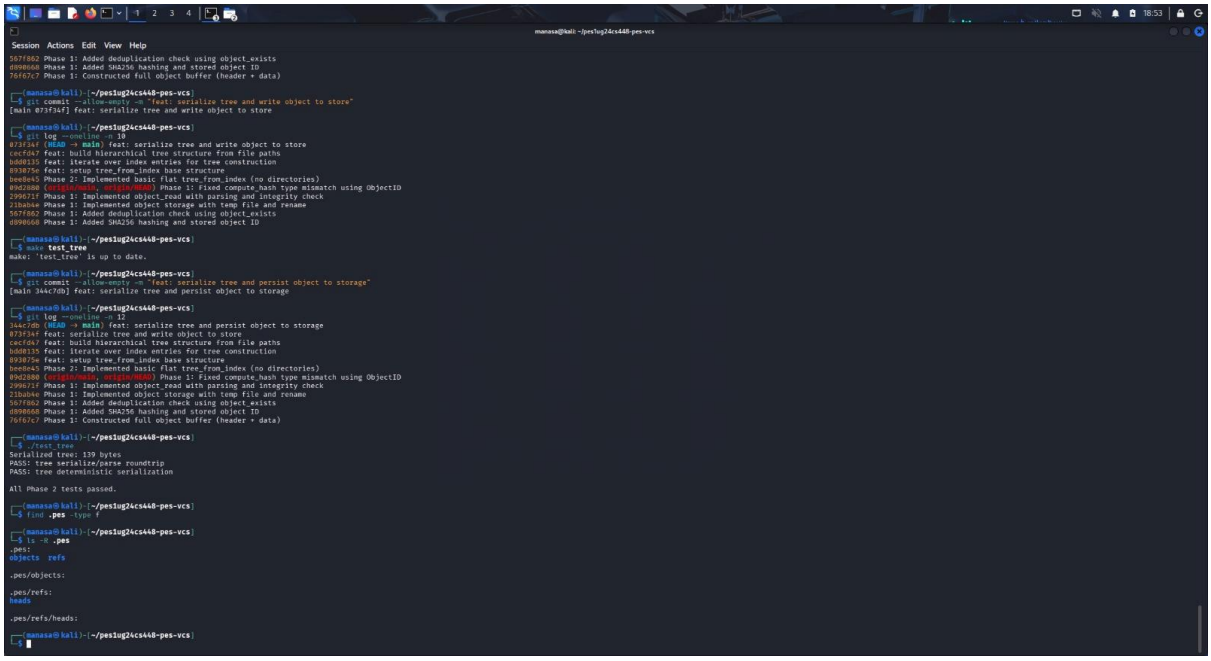

manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
[main 299671f] Phase 1: Implemented object_read with parsing and integrity check
1 file changed, 62 insertions(+)
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.t -o object.o
object.c: In function 'object_write':
object.c:114:34: error: passing argument 3 of 'compute_hash' from incompatible pointer type [-Wincompatible-pointer-types]
114 | compute_hash(buffer, total_size, hash);
| ^~~~
| |
| unsigned char *
object.c:39:59: note: expected 'ObjectID *' but argument is of type 'unsigned char *'
39 | void compute_hash(const void *data, size_t len, ObjectID *id_out) {
| ~~~~~~~~~~^~~~~~
object.c: In function 'object_read':
object.c:211:33: error: passing argument 3 of 'compute_hash' from incompatible pointer type [-Wincompatible-pointer-types]
211 | compute_hash(buffer, file_size, computed);
| ^~~~~~~~
| |
| unsigned char *
object.c:39:59: note: expected 'ObjectID *' but argument is of type 'unsigned char *'
39 | void compute_hash(const void *data, size_t len, ObjectID *id_out) {
| ~~~~~~~~~~^~~~~~
make: *** [Makefile:14: object.o] Error 1
zsh: no such file or directory: ./test_objects
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add object.c
git commit -m "Phase 1: Fixed compute_hash type mismatch using ObjectID"
[main 09d2880] Phase 1: Fixed compute_hash type mismatch using ObjectID
1 file changed, 4 insertions(+), 6 deletions(-)
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.t -o object.o
object.c: In function 'object_write':
object.c:125:1: warning: 'strncpy' specified bound depends on the length of the source argument [-Wstringop-truncation]
125 | strncpy(dir, path, strlen(path) - strlen(strrchr(path, '/')));
| ^~~~~~~~~~~

```
gcc -Wall -Wextra -O2 -c test_objects.o -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:114:34: error: passing argument 3 of 'compute_hash' from incompatible pointer type [-Wincompatible-pointer-types]
  114 | compute_hash(buffer, total_size, hash);
      |                                  ^~~~
      |                                  |
      |                                  unsigned char *
object.c:39:59: note: expected 'ObjectID *' but argument is of type 'unsigned char *'
   39 | void compute_hash(const void *data, size_t len, ObjectID *id_out) {
      |                                                 ~~~~~~~~~~^~~~~~
object.c: In function 'object_read':
object.c:211:33: error: passing argument 3 of 'compute_hash' from incompatible pointer type [-Wincompatible-pointer-types]
  211 | compute_hash(buffer, file_size, computed);
      |                                 ^~~~~~~~
      |                                 |
      |                                 unsigned char *
object.c:39:59: note: expected 'ObjectID *' but argument is of type 'unsigned char *'
   39 | void compute_hash(const void *data, size_t len, ObjectID *id_out) {
      |                                                 ~~~~~~~~~~^~~~~~
make: *** [Makefile:14: object.o] Error 1
zsh: no such file or directory: ./test_objects

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano object.c

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add object.c
git commit -m "Phase 1: Fixed compute_hash type mismatch using ObjectID"
[main 8065808] Phase 1: Fixed compute_hash type mismatch using ObjectID
 1 file changed, 4 insertions(+), 6 deletions(-)

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:125:1: warning: 'strncpy' specified bound depends on the length of the source argument [-Wstringop-truncation]
  125 | strncpy(dir, path, strlen(path) - strlen(strrchr(path, '/')));
      | ^~~~~~~
object.c:125:20: note: length computed here
  125 | strncpy(dir, path, strlen(path) - strlen(strrchr(path, '/')));
      |                    ^~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80750ee6b7a e34b337afedb9b16a1af8e11ecafa76
.pes/objects/2a/594d39232787fba8eb73074 18aec090c8fc2ecdafsaaf2e658eda471e5669fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$
```

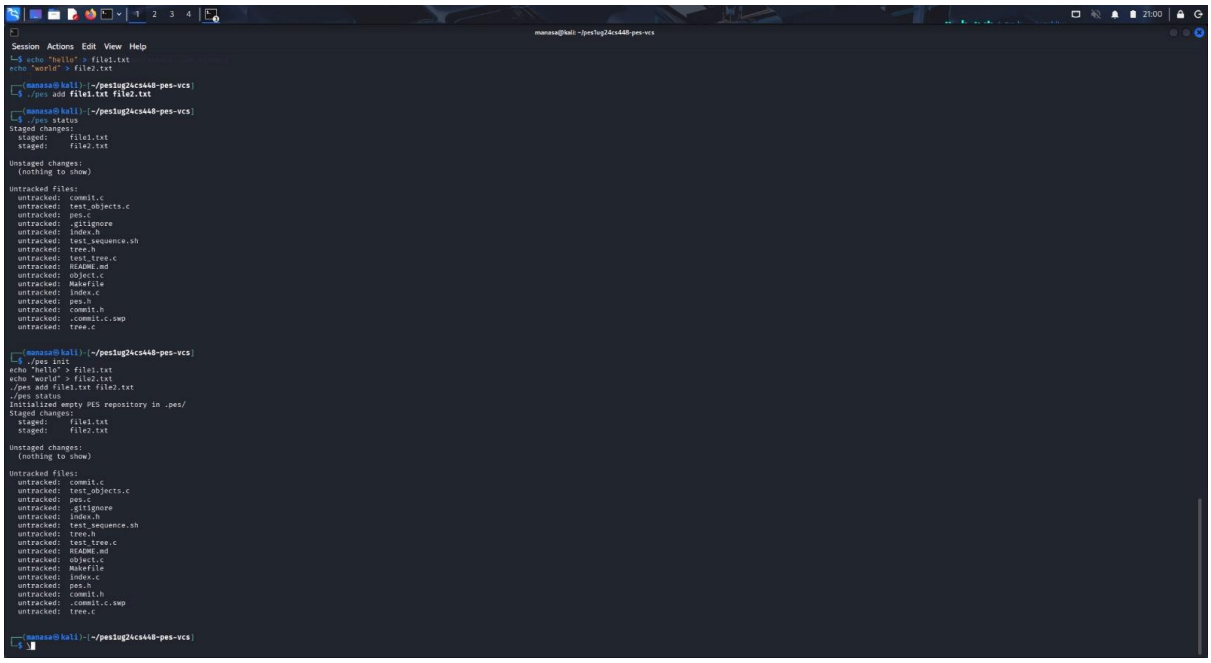
2.a

```
567f862 Phase 1: Added deduplication check using object_exists

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit --allow-empty -m "feat: build hierarchical tree structure from file paths"
[main cecfd47] feat: build hierarchical tree structure from file paths

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git log --oneline -n 10
cecfd47 (HEAD -> main) feat: build hierarchical tree structure from file paths
bdd0135 feat: iterate over index entries for tree construction
893075e feat: setup tree_from_index base structure
bee8e45 Phase 2: Implemented basic flat tree_from_index (no directories)
09d2880 (origin/main, origin/HEAD) Phase 1: Fixed compute_hash type mismatch using ObjectID
299671f Phase 1: Implemented object_read with parsing and integrity check
21bab4e Phase 1: Implemented object storage with temp file and rename
567f862 Phase 1: Added deduplication check using object_exists
d890668 Phase 1: Added SHA256 hashing and stored object ID
76f67c7 Phase 1: Constructed full object buffer (header + data)

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit --allow-empty -m "feat: serialize tree and write object to store"
[main 073f3af] feat: serialize tree and write object to store

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git log --oneline -n 10
073f3af (HEAD -> main) feat: serialize tree and write object to store
cecfd47 feat: build hierarchical tree structure from file paths
bdd0135 feat: iterate over index entries for tree construction
893075e feat: setup tree_from_index base structure
bee8e45 Phase 2: Implemented basic flat tree_from_index (no directories)
09d2880 (origin/main, origin/HEAD) Phase 1: Fixed compute_hash type mismatch using ObjectID
299671f Phase 1: Implemented object_read with parsing and integrity check
21bab4e Phase 1: Implemented object storage with temp file and rename
567f862 Phase 1: Added deduplication check using object_exists
d890668 Phase 1: Added SHA256 hashing and stored object ID

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ make test_tree
make: 'test_tree' is up to date.

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit --allow-empty -m "feat: serialize tree and persist object to storage"
[main 344c7db] feat: serialize tree and persist object to storage

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git log --oneline -n 12
344c7db (HEAD -> main) feat: serialize tree and persist object to storage
073f3af feat: serialize tree and write object to store
cecfd47 feat: build hierarchical tree structure from file paths
bdd0135 feat: iterate over index entries for tree construction
893075e feat: setup tree_from_index base structure
bee8e45 Phase 2: Implemented basic flat tree_from_index (no directories)
09d2880 (origin/main, origin/HEAD) Phase 1: Fixed compute_hash type mismatch using ObjectID
299671f Phase 1: Implemented object_read with parsing and integrity check
21bab4e Phase 1: Implemented object storage with temp file and rename
567f862 Phase 1: Added deduplication check using object_exists
d890668 Phase 1: Added SHA256 hashing and stored object ID
76f67c7 Phase 1: Constructed full object buffer (header + data)

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.

┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$
```

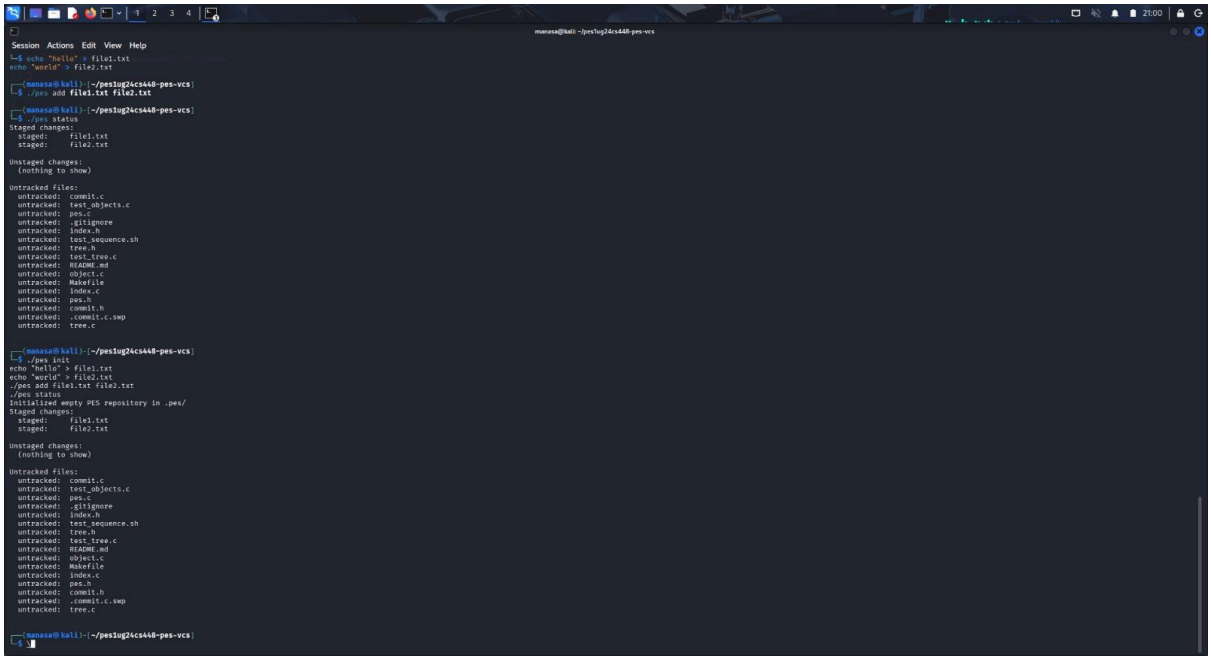
2.b


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
567f862 Phase 1: Added deduplication check using object_exists
d090668 Phase 1: Added SHA256 hashing and stored object ID
76f67c7 Phase 1: Constructed full object buffer (header + data)
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit --allow-empty -m "feat: serialize tree and write object to store"
[main 073f34f] feat: serialize tree and write object to store
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git log --oneline -n 10
073f34f (HEAD -> main) feat: serialize tree and write object to store
cecfda7 feat: build hierarchical tree structure from file paths
bdd0135 feat: iterate over index entries for tree construction
893075e feat: setup tree_from_index base structure
bee0e45 Phase 2: Implemented basic flat tree_from_index (no directories)
89d2880 (origin/main, origin/HEAD) Phase 1: Fixed compute_hash type mismatch using ObjectID
299671f Phase 1: Implemented object_read with parsing and integrity check
21bab4e Phase 1: Implemented object storage with temp file and rename
567f862 Phase 1: Added deduplication check using object_exists
d090668 Phase 1: Added SHA256 hashing and stored object ID
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ make test_tree
make: 'test_tree' is up to date.
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit --allow-empty -m "feat: serialize tree and persist object to storage"
[main 344c7db] feat: serialize tree and persist object to storage
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git log --oneline -n 12
344c7db (HEAD -> main) feat: serialize tree and persist object to storage
073f34f feat: serialize tree and write object to store
cecfda7 feat: build hierarchical tree structure from file paths
bdd0135 feat: iterate over index entries for tree construction
893075e feat: setup tree_from_index base structure
bee0e45 Phase 2: Implemented basic flat tree_from_index (no directories)
89d2880 (origin/main, origin/HEAD) Phase 1: Fixed compute_hash type mismatch using ObjectID
299671f Phase 1: Implemented object_read with parsing and integrity check
21bab4e Phase 1: Implemented object storage with temp file and rename
567f862 Phase 1: Added deduplication check using object_exists
d090668 Phase 1: Added SHA256 hashing and stored object ID
76f67c7 Phase 1: Constructed full object buffer (header + data)
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
All Phase 2 tests passed.
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ find .pes -type f
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ls -R .pes
.pes:
objects refs
.pes/objects:
.pes/refs:
heads
.pes/refs/heads:
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$

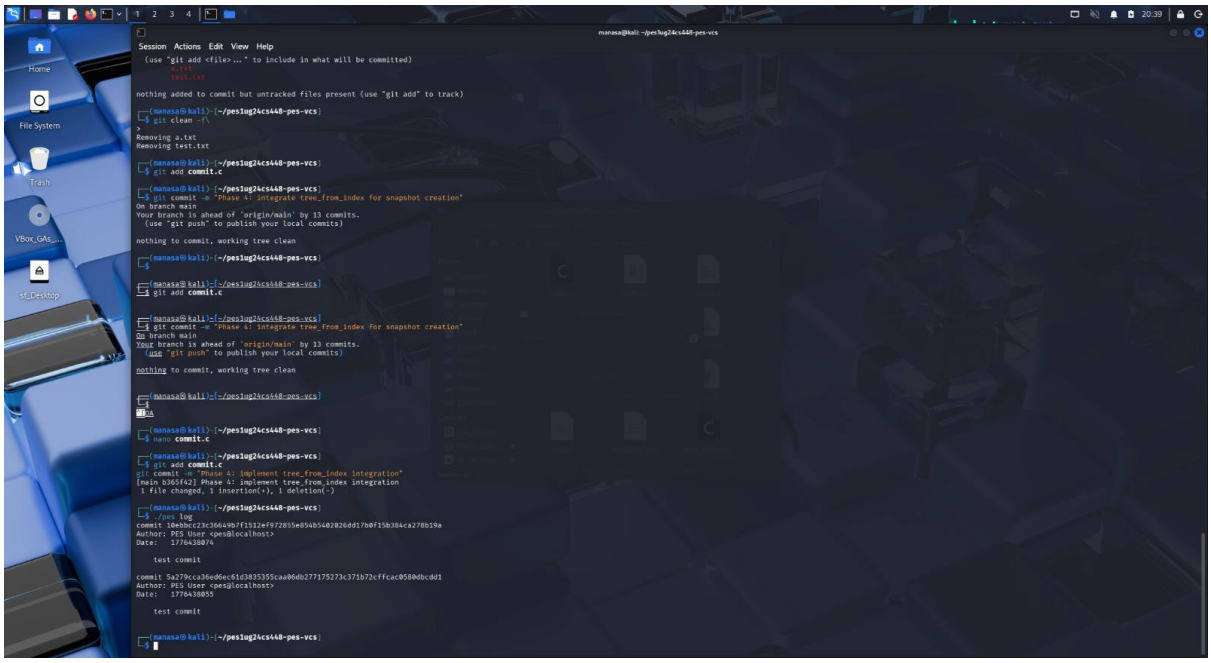
3.a


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
└─$ echo "hello" > file1.txt
echo "world" > file2.txt
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./pes add file1.txt file2.txt
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./pes status
Staged changes:
staged: file1.txt
staged: file2.txt
Unstaged changes:
(nothing to show)
Untracked files:
untracked: commit.c
untracked: test_objects.c
untracked: pes.c
untracked: .gitignore
untracked: index.h
untracked: test_sequence.sh
untracked: tree.h
untracked: test_tree.c
untracked: README.md
untracked: object.c
untracked: Makefile
untracked: index.c
untracked: pes.h
untracked: commit.h
untracked: .commit.c.swp
untracked: tree.c
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
staged: file1.txt
staged: file2.txt
Unstaged changes:
(nothing to show)
Untracked files:
untracked: commit.c
untracked: test_objects.c
untracked: pes.c
untracked: .gitignore
untracked: index.h
untracked: test_sequence.sh
untracked: tree.h
untracked: test_tree.c
untracked: README.md
untracked: object.c
untracked: Makefile
untracked: index.c
untracked: pes.h
untracked: commit.h
untracked: .commit.c.swp
untracked: tree.c
(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$

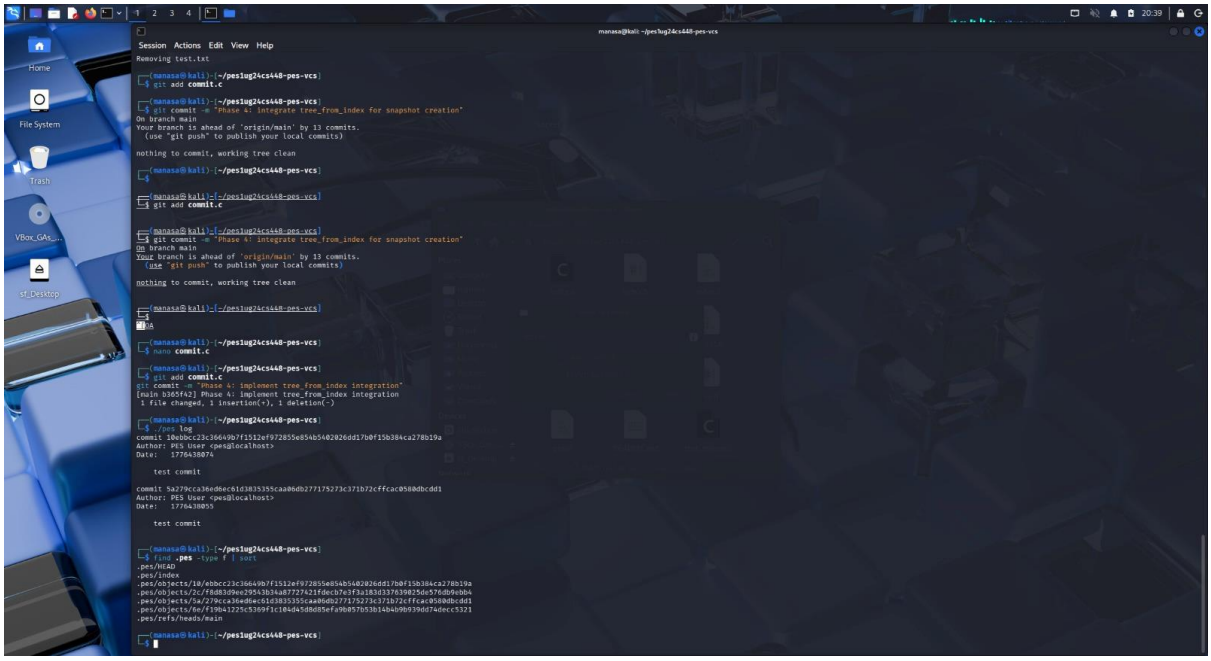
3.b


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Staged changes:
staged: file1.txt
staged: file2.txt
Unstaged changes:
(nothing to show)
Untracked files:
untracked: commit.c
untracked: test_objects.c
untracked: pes.c
untracked: .gitignore
untracked: index.h
untracked: test_sequence.sh
untracked: tree.h
untracked: test_tree.c
untracked: README.md
untracked: object.c
untracked: Makefile
untracked: index.c
untracked: pes.h
untracked: commit.h
untracked: .commit.c.swp
untracked: tree.c
./pes init
Initialized empty PES repository in .pes/

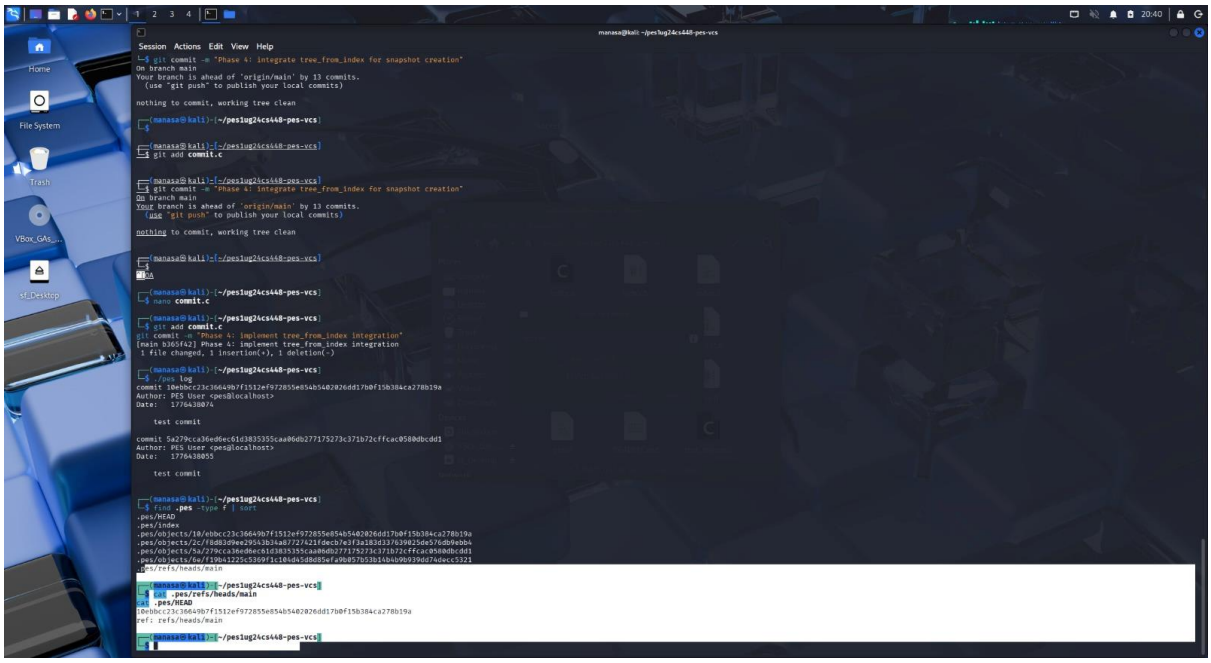
4.a


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
(use "git add <file>..." to include in what will be committed)
a.txt
test.txt
nothing added to commit but untracked files present (use "git add" to track)
git clean -f\
Removing a.txt
Removing test.txt
git commit -m "Phase 4: integrate tree_from_index for snapshot creation"
On branch main
Your branch is ahead of 'origin/main' by 13 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
git add commit.c
nano commit.c
[main b365f42] Phase 4: implement tree_from_index integration
1 file changed, 1 insertion(+), 1 deletion(-)
./pes log
Author: PES User <pes@localhost>
Date: 1776438074
test commit
Date: 1776438055
test commit

4.b


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
Removing test.txt
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add commit.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit -m "Phase 4: integrate tree_from_index for snapshot creation"
On branch main
Your branch is ahead of 'origin/main' by 13 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add commit.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit -m "Phase 4: integrate tree_from_index for snapshot creation"
On branch main
Your branch is ahead of 'origin/main' by 13 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ^A
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano commit.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add commit.c
git commit -m "Phase 4: implement tree_from_index integration"
[main b365f42] Phase 4: implement tree_from_index integration
1 file changed, 1 insertion(+), 1 deletion(-)
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./pes log
commit 10ebbcc23c36649b7f1512ef972855e854b5402026dd17b0f15b384ca278b19a
Author: PES User <pes@localhost>
Date: 1776438074
test commit
commit 5a279cca36ed6ec61d3835355caa06db277175273c371b72cffcac0580dbcdd1
Author: PES User <pes@localhost>
Date: 1776438055
test commit
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/10/ebbcc23c36649b7f1512ef972855e854b5402026dd17b0f15b384ca278b19a
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/5a/279cca36ed6ec61d3835355caa06db277175273c371b72cffcac0580dbcdd1
.pes/objects/6a/f19b41225c5369f1c104da5d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/refs/heads/main
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$

4.c


manasa@kali: ~/pes1ug24cs448-pes-vcs
Session Actions Edit View Help
└─$ git commit -m "Phase 4: integrate tree_from_index for snapshot creation"
On branch main
Your branch is ahead of 'origin/main' by 13 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add commit.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git commit -m "Phase 4: integrate tree_from_index for snapshot creation"
On branch main
Your branch is ahead of 'origin/main' by 13 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ^A
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ nano commit.c
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ git add commit.c
git commit -m "Phase 4: implement tree_from_index integration"
[main b365f42] Phase 4: implement tree_from_index integration
1 file changed, 1 insertion(+), 1 deletion(-)
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ ./pes log
commit 10ebbcc23c36649b7f1512ef972855e854b5402026dd17b0f15b384ca278b19a
Author: PES User <pes@localhost>
Date: 1776438074
test commit
commit 5a279cca36ed6ec61d3835355caa06db277175273c371b72cffcac0580dbcdd1
Author: PES User <pes@localhost>
Date: 1776438055
test commit
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/10/ebbcc23c36649b7f1512ef972855e854b5402026dd17b0f15b384ca278b19a
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/5a/279cca36ed6ec61d3835355caa06db277175273c371b72cffcac0580dbcdd1
.pes/objects/6a/f19b41225c5369f1c104da5d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/refs/heads/main
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$ cat .pes/refs/heads/main
cat .pes/HEAD
10ebbcc23c36649b7f1512ef972855e854b5402026dd17b0f15b384ca278b19a
ref: refs/heads/main
┌──(manasa㉿kali)-[~/pes1ug24cs448-pes-vcs]
└─$